# 1
# Introduction

## 1.1 Objectives

As a result of studying this chapter, the student should be able to:

- Describe the three different types of processes;
- Indicate the meaning of a Time constant;
- Describe the meaning of Process Variable, Set Point and Output;
- List the different modes of operation of a control system.

## 1.2 Introduction

In principle most process control systems consist of a control loop, comprising four main functions, these being:

- A measurement of the state or condition of a Process;
- A Controller calculating an action based on this measured value against a pre-set or control value (Set Point);
- A signal with a value that represents the result of this calculation being fed back from the controller to the process, which is to manipulate the process action;
- The process itself reacting to this signal, and changing its state or condition.

Unfortunately, it is only within the scope of this manual for us to consider the first three component parts, the measurement of a process, control calculations based on this measurement and how the final result is used. Every process is different, and thereby hangs a problem, the object we are striving to achieve is control of the process, and that places us somewhat in a dilemma as we do not know what that process is, but we cannot ignore it.

In addition to this is a second problem of knowing where to start when the entire process is a loop. Do we start with measurement, or the controller or what? Fortunately, a solution to the first problem, that of unknown process, gives us the solution to the second problem.

All process must have a set of common parameters and dynamics; if they did not then every type of controller would have to be different and no common boundaries would exist.

The dynamics of every process, their type and magnitude, have to be understood before any attempt can be made in selecting the type of measuring device(s), the type of control system and finally the type of final control element.

Let us then start by examining the component parts of the more important dynamics that are common to all process functions. This will be the topic covered in the next few sections of this chapter, and upon completion we should be able to draw a simple block for any process system, saying for example, "It is a system with Capacitive and Inertial properties, and as such we can expect it to perform in the following way", regardless of what or how the process is manufacturing its final product.

From this analytical result an accurate selection of the type of measuring transducer can be selected, this being covered in chapter two and likewise the selection of the final control element can be correctly selected, this being covered in chapter three.

From thereon Chapters 4 through 14 deal with all the other aspects of Practical Process Control, namely the controller(s), functions, actions and reactions, function combinations and various modes of operation. By way of introduction to the controller itself, the last sections of this chapter are introductions to the basic definitions of controller terms and types of control modes that are available.

## 1.3 Process Dynamics

In order to match a controller to a process it is necessary to understand something of the process. The majority of processes contain resistance, capacitance and dead-time elements, which will determine the dynamic and steady-state responses of the process to disturbances.

To visualize the behavior of a system, block diagrams can be used to provide a visual description of the components. The two main symbols used are the circle and the rectangular block as shown in Figure 1.1.

The circle represents algebraic addition and subtraction. It is always entered by two lines but exited by only one line.

The rectangular block represents algebraic multiplication and division. It is entered and exited by only one line. The output is the product of the system function, which is symbolized inside the block, with the input.
The system function is the symbolic representation of how an input change affects the output for a particular process component.

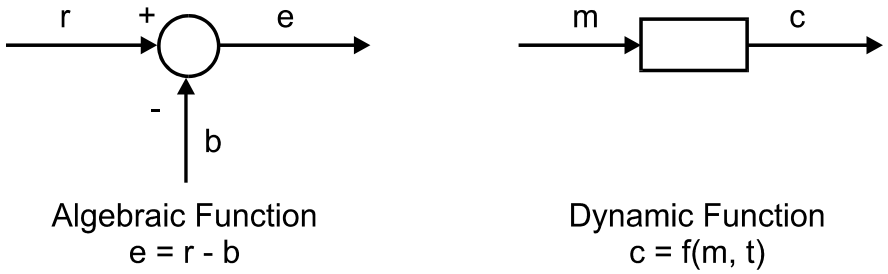



***Figure 1.1***
***Diagram of the Summer and Gain Blocks***

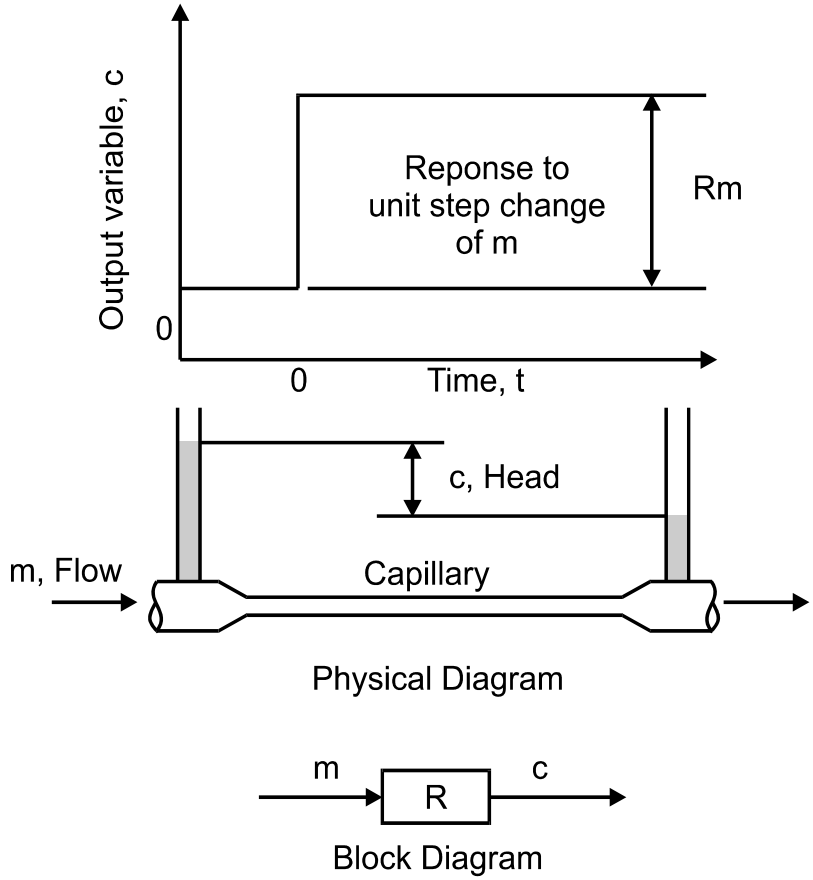
### 1.3.1 Resistance Type Processes

The most obvious illustration of a resistance type process is the pressure drop through pipes and other equipment, i.e. where there is some resistance to the transfer of energy or mass. These parts of the process are known as 'resistances'.

Figure 1.2 illustrates the operation of a capillary flow system where the flow is linearly proportional to the pressure drop. This process is described by a steady state gain, equal to the resistance (R). As the input (flow = m) changes instantaneously from zero to m, the output (head = c) undergoes an instantaneous change from zero to c = R x m.

This laminar resistance to flow is analogous to the electrical resistance to current flow.

In laminar flow, such as capillary flow, the resistance is a function of the square root of the pressure drop.




$$c = Rm$$

where c = output resistance (head)
R = resistance
m = input variable (flow)

***Figure 1.2***
***A capillary flow system illustrating the resistance (Proportional) Element.***

Flow processes usually consists of a flow measuring device and a control valve in series with it.; with the flow (c) passing through both of them (Figure 1.3).

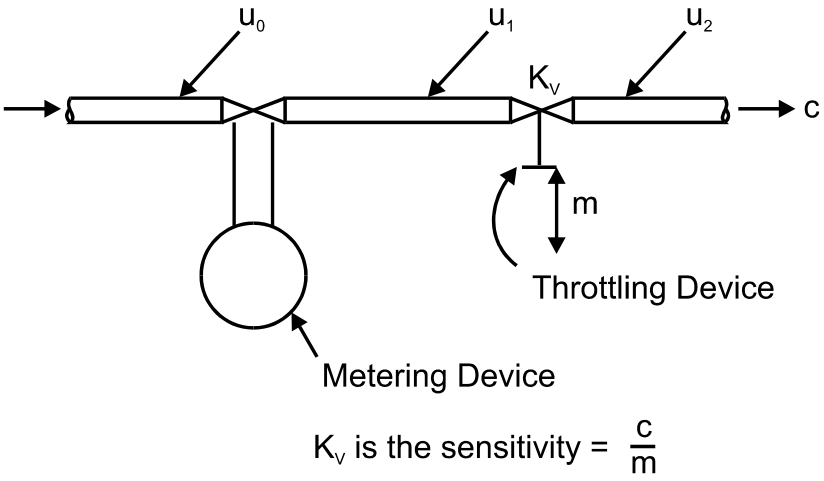
The block diagram illustrates that this is an algebraic and proportional (resistive) process.

The manipulated variable (m) is the operation of the control valve, and (c) is the controlled variable; this being the flow through the system. A change in (m) results in an immediate and proportional change in (c).

The amount of change is a function of process gain or sensitivity $K_A$. Load variables are $U_0$ and $U_2$ the up and down stream pressures and any change of these results in an immediate and proportional change in the flow (c). The amount of change is proportional to their process sensitivity ($K_B$).

The overall process equation is

$$c = (K_A)m + (K_B)U_O - (K_B)U_2 + M_3$$



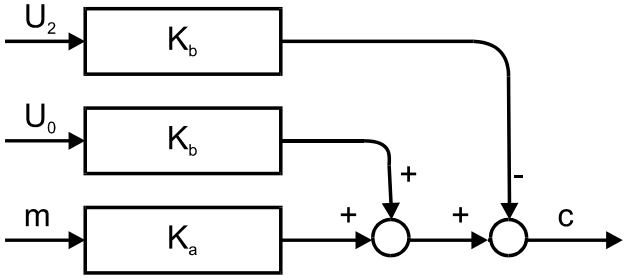



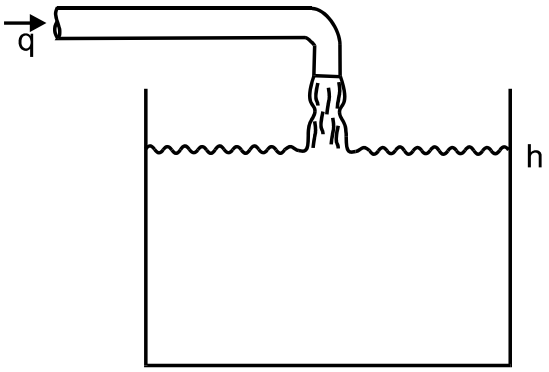

***Figure 1.3***
***Flow is a resistance (Proportional) process***

## 1.3.2 Capacitance Type Processes

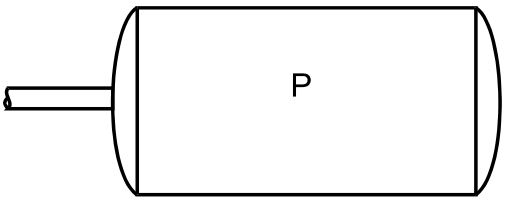
Most processes include some form of capacitance or storage capability, either for materials (gas, liquid, or solids) or for energy (thermal, chemical, etc.). Those parts of the process with the ability to store mass or energy are termed 'capacities'. Thermal capacitance is directly analogous to electrical capacitance.

The capacitance of a liquid or gas storage tank is expressed in area units. These processes are illustrated in Figure 1.4. The gas capacitance of a tank is constant and is analogous to electrical capacitance.

The liquid capacitance equals the cross-sectional area of the tank at the liquid surface; if this is constant then the capacitance is also constant at any head.




Liquid capacitance is defined by $C = \frac{dv}{dh}$ $ft^2$



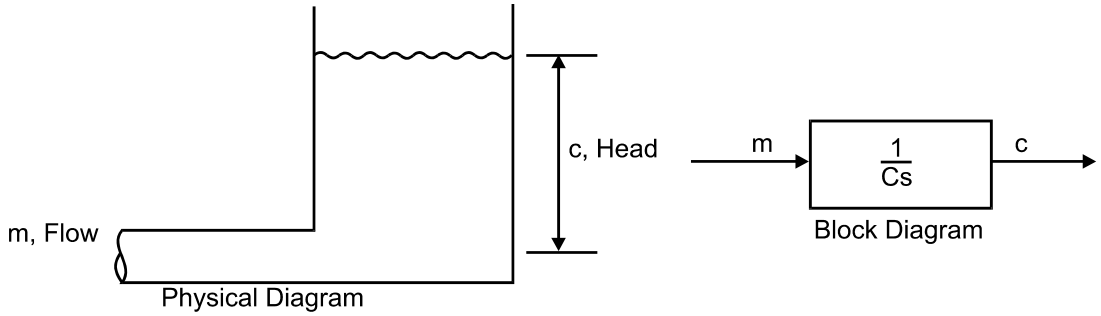

C, Capacitance

Gas capacitance is defined by

$C = \frac{dv}{dp} = \frac{V}{nRT} ft^2$

where v = weight of gas in vessel, lb.
V = volume of vessel, $ft^3$
R = Gas constant of a specific gas ft/deg
p = pressure, $lb.ft^2$
n = polytropic exponent is between 1.0 and 1 .2 for uninsulated tanks

***Figure 1.4***
***Capacitance of a liquid or gas storage tank expressed in area units.***




$$C\frac{dc}{dt} = m = (Cs)c = m \therefore c = \left(\frac{1}{Cs}\right)m$$

where C = capacitance
c = output variable (head)
t = time
m = input variable (flow)
$s = \frac{d}{dt}$ = differential operator

***Figure 1.5***
***Liquid capacitance Calculation; the Capacitance Element.***

A purely capacitive process element can be illustrated by a tank with only an inflow connection such as Figure 1.5. In such a process, the rate at which the level rises is inversely proportional to the capacitance and the tank will eventually flood. For an initially empty tank with constant inflow, the level c is the product of the inflow rate m and the time period of charging t divided by the capacitance of the tank C.

### 1.3.3 Inertia Type Processes

Inertia effects are due to the motion of matter. They are most commonly associated with mechanical systems involving moving components, but are also important in some flow systems in which fluids must be accelerated or decelerated.

Resistance and capacitance are perhaps the most important effects in industrial processes involving heat transfer, mass transfer, and fluid flow operations. The combined effect of supplying a capacity through a resistance is a time retardation. This is basic to most dynamic systems found in industrial processes.

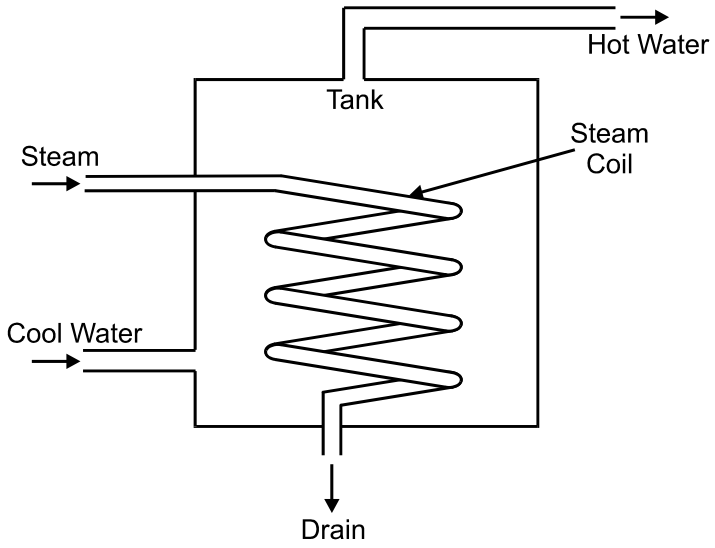



***Figure 1.6***
***Resistance and Capacitance Effects in a water Heater***

As a result of this time retardation, an instantaneous change in the input to the system will not result in an instantaneous change in the output. Rather, the response will be slow, requiring a finite period of time to attain a new equilibrium.

## 1.4 Process Time Constants

Combining a capacitance type process element, such as a tank, and a resistance type process element, such as a valve, results in a single time constant process. In a mathematical sense, the time constant is the future time necessary to experience 63.2% of the change remaining to occur, at any moment in the process.

The time constant is a measure of the rapidity of the response of the process. It can be characterized in terms of the capacitance and resistance (or conductance) of the process.

### 1.4.1 First Order Response

In the basic case of a first order response, the maximum rate of change of output occurs immediately after a step change in input. 63.2% of the total response is attained after one time constant. If the system continued to change at its initial speed of response, the maximum response rate, it would reach 100% of the output change in one time constant. A physical example of a first order process is an initially empty tank with a constant inflow and a valve controlled outflow.

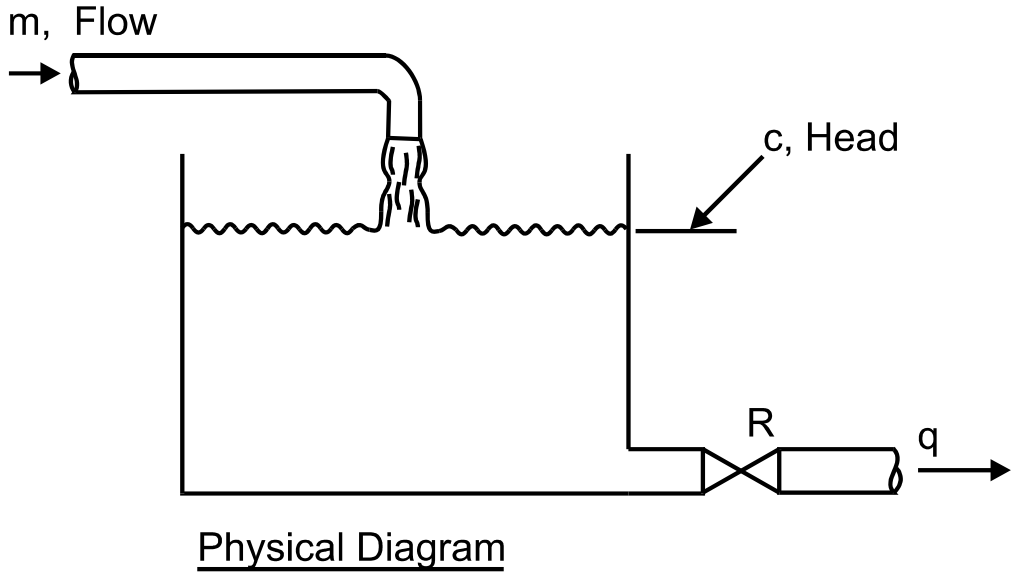



***Figure 1.7***
***A physical example of a first order process a constant inflow and a valve controlled outflow.***

The general equation for such a process is a linear first order differential equation:

$$T\frac{dc(t)}{dt} + c(t) = Km(t)$$

where:

| | |
|---|---|
| T, K | = constants of the process, time constant and gain |
| t | = time |
| c(t) | = process output response |
| m(t) | = process input response |

Process elements of this description are common and are generally referred to as first order lags. They may also be called linear lags or exponential transfer lags. Components with this response are characterized by the capacity to store material or energy and the dynamic shape of the response curve is described by a time constant.

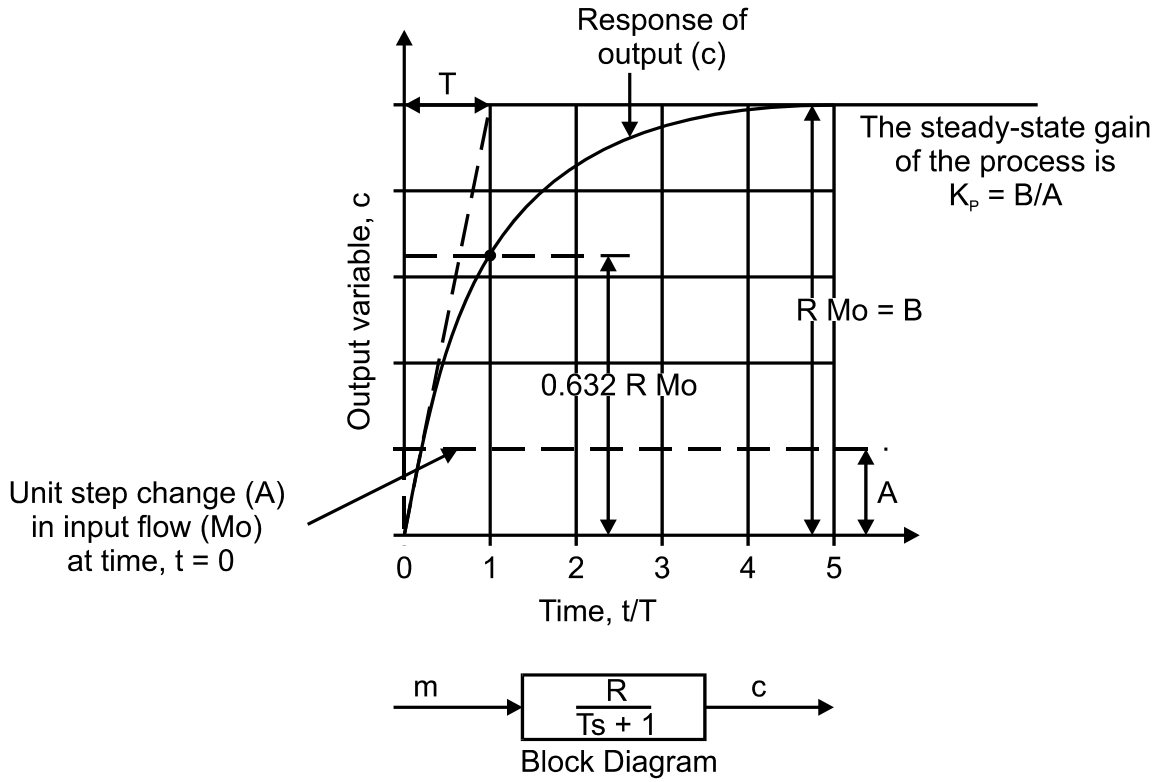



***Figure 1.8***
***First Order Response***

In multiple time constant processes, say where two tanks are connected in series, the system therefore has two time constants operating in series. As the number of time constants increases, the response curves of the system become progressively more retarded and the overall response gradually changes into an S-shaped reaction curve. This curve is typical of the majority of processes.

### 1.4.2 Second Order Response

Second order processes result in a more complicated response curve. This is due to inertia effects and interactions between first order resistance and capacitance elements. They are described by the second order differential equation:

$$\frac{d^2}{dt^2} c(t) + 2x w_n \frac{d}{dt} c(t) + w^2_n c(t) = K w^2_n r(t)$$

Where:

| | |
|---|---|
| $w_n$ | = the natural frequency of the system |
| x | = the damping ratio of the system |
| K | = the system gain |
| t | = time |
| r(t) | = input response of the system |
| c(t) | = output response of the system |

The solutions to the equation for a step change in r(t) with all initial conditions zero can be any one of a family of curves shown in Figure 1.9. There are three possible cases in the solution, depending on the value of the damping ratio:

x < 1.0, the system is underdamped and overshoots the steady-state value.

If x < 0.707, the system will oscillate about the final steady-state value.

x > 1.0, the system is overdamped and will not oscillate or overshoot the final steady-state value.

x = 1.0, the system is critically damped. In this state it yields the fastest response without overshoot or oscillation.

The natural frequency term $w_n$ is related to the speed of the response for a particular value of x. It is defined in terms of the 'perfect' or 'frictionless' situation where x = 0.0.

A large frequency tends to squeeze the response and a small frequency to stretch it.

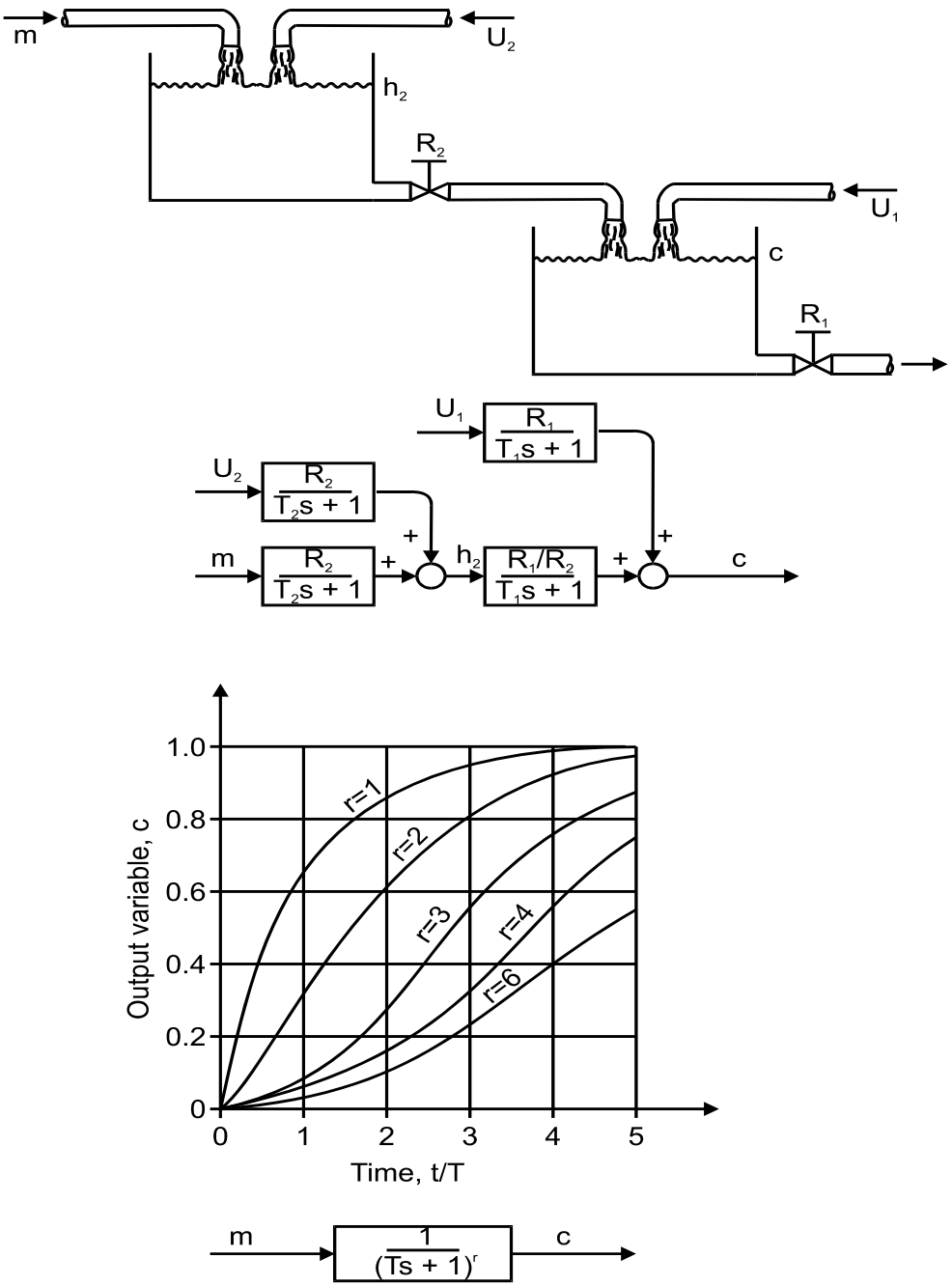



***Figure 1.9***
***Second Order Response; Two time constants in series, and response curves of processes with several time constants.***

### 1.4.3 High Order Response

Time delays can be used to approximate high-order model dynamics. Any process that consists of a large number of process units connected in series, can be represented by a high order response. This transfer function could represent a series of first order transfer functions. In practice, the mathematical analysis of uncontrolled processes containing time delays is relatively simple but a time delay, or a set of time delays, within a feedback loop tends to lend itself to very complex mathematics.

In general, the presence of time delays in control systems reduces the effectiveness of the controller. In well-designed systems the time delays (deadtimes) should be kept to the minimum.

## 1.5 Basic Definitions and Terms used in Process Control

As we will see in chapters two and three, two of the most important signals used in process control are called

PROCESS VARIABLE or PV.

and the

MANIPULATED VARIABLE or MV.

In industrial process control, the PROCESS VARIABLE or PV is measured by an instrument in the field and acts as an input to a computer based automatic controller which takes action based on the value of it. Alternatively the PV can be an input to computer based hardware system and its value displayed in some manner so that the operator can perform manual control and supervision.

The variable to be manipulated, in order to have control over the PV, is called the MANIPULATED VARIABLE or MV. If we control a particular flow for instance, we manipulate a valve to control the flow. Here, the valve position is called the MANIPULATED VARIABLE and the measured flow becomes the PROCESS VARIABLE.

In the case of a simple automatic controller, the Controller OUTPUT Signal (OP) drives the Manipulated Variable. In more complex automatic control systems, a controller output signal may not always drive a Manipulated Variable in the field. In practice, the term Manipulated Variable is rarely used. Most people involved in process control refer to the OP (output) of a controller and it is assumed that one knows the purpose of it.

The ideal value of the PV is often called TARGET VALUE. and in the case of an automatic control, the term SET POINT VALUE is preferred.

## 1.6 Types or Modes of Operation of Available Control Systems

There are five basic forms of control available in Process Control. These are:

- On-Off
- Modulating
- Open Loop
- Feed Forward
- Closed loop

The next five sections; 1.4.1 to 1.4.5; examine each of these in turn.

### 1.6.1 On-Off Control

The most basic control concept is ON-OFF Control as found in a modern iron in our households. This is a very crude form of control, which nevertheless should be considered as a cheap and effective means of control if a fairly large fluctuation of the PV (Process Variable) is acceptable.

The wear and tear of the controlling element (solenoid valve etc) needs special consideration. As the bandwidth of fluctuation of a PV is increased, the frequency of switching (and thus wear and tear) of the controlling element decreases.

### 1.6.2 Modulating Control

If the output of a controller can move through a range of values, we have modulating control. It is understood that modulating control takes place within a defined operating range (with an upper and lower limit) only.

Modulating control can be used in both open and closed loop control systems.

### 1.6.3 Open Loop Control

We have open loop control, if the control action (Controller Output Signal OP) is not a function of the PV (Process Variable) or load changes. The open loop control does not self-correct, when these PV's drift.

### 1.6.4 Feed Forward Control

Very often it is a form of control based on measured disturbances (Feed Forward Control). It is a form of Open Loop control as the PV is not used in the control action.

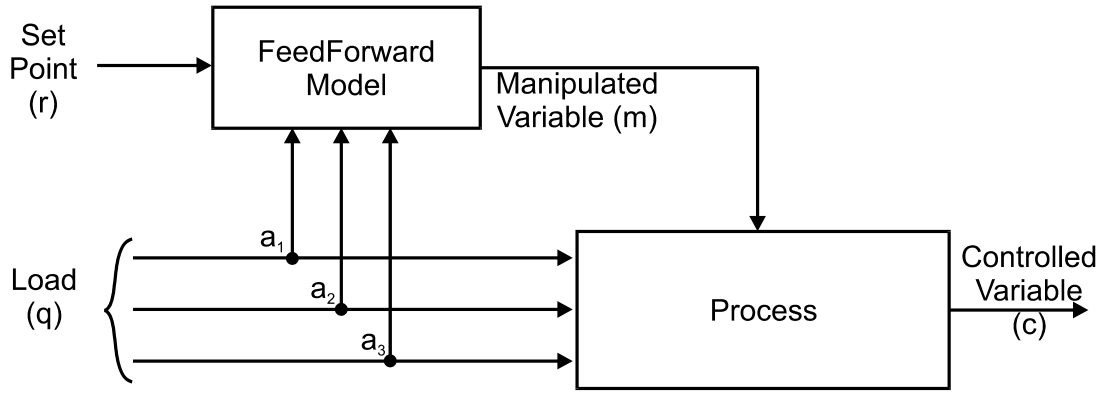



***Figure 1.10***
***The Feedforward Control Loop***

Feedforward is a more direct form of control than finding the correct value of the manipulated variable (MV) by trial and error as occurs in feedback control. In feedforward the major process variables are fed into a model to calculate the Manipulated Variable (MV) required to control at the Set Point (SP).

Figure 1.10 shows the principle of an Open Loop Control block. The PV (Controlled Variable (c) ) is just a result of the control action.

The proper industrial use of this type of control is, for example, to incorporate this with feedback; or closed loop control (section 1.6.5) . Then the imperfect feedforward control can correct up to 90% of the upsets, leaving the feedback system to correct the 10% bias left by the feedforward component.

### 1.6.5 Closed Loop or Feedback Control

We have a Closed Loop Control System if the PV, the objective of control, is used to determine the control action. The principle is shown below in Figure 1.11.

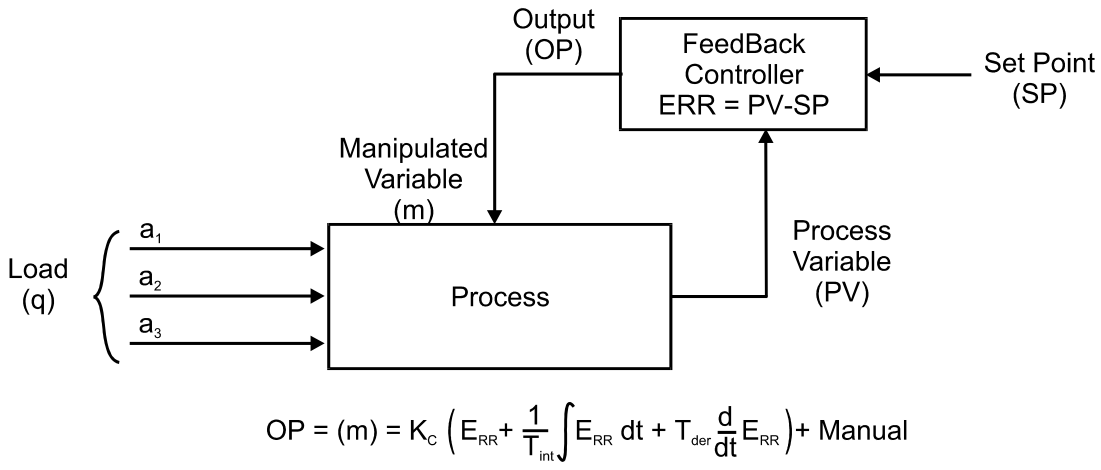



***Figure 1.11***
***The Feedback Control Loop***

The idea of Closed Loop Control is to measure the PV (Process Variable); compare this with the SP (Setpoint), which is the desired or target value; and determine a control action which results in a change of the OP (Output) value of an automatic controller.

In most cases, the ERROR (ERR) term is used to calculate the OP value.

ERR = PV - SP

If ERR = SP -PV has to be used, the controller has to be set for REVERSE control action.

## 1.7 Closed Loop Controller and Process Gain Calculations

Within this closed loop form of control there are two functional gain blocks, one being in the controller and the other in the process being controlled. The LOOP gain (KLOOP) is the product of the CONTROLLER GAIN ($K_C$) and the PROCESS GAIN ($K_P$).

$$\text{LOOPGAIN } (K_{LOOP}) = K_C \times K_P = \frac{\Delta MV}{\Delta E} X \frac{\Delta PV}{\Delta MV} = \frac{\Delta PV}{\Delta E}$$

where:

$$\text{Process Gain } (K_P) = \frac{\Delta PV}{\Delta MV}$$

and

$$\text{Controller Gain } (K_C) = \frac{\Delta MV}{\Delta E}$$

As the total constituent parts of the entire loop consist of a minimum of 4 functional items; the Process Gain $(K_P) = \frac{\Delta PV}{\Delta MV}$ Controller Gain $(K_C) = \frac{\Delta MV}{\Delta E}$ the measuring transducer or Sensor gain (refer to chapter 2), $K_S$ and finally the Valve Gain $K_V$ the total loop gain is the product of these four operational blocks.

For 1/4 damping, the ideal response where each oscillation has a 1/4 of the amplitude of the previous cycle then:

$$\text{KLOOP} = (K_C \times K_P) = \left(\frac{\Delta MV}{\Delta E} X \frac{\Delta PV}{\Delta MV} = \frac{\Delta PV}{\Delta E}\right) \text{ x } K_S \text{ x } K_V = 0.5$$

## 1.8 Proportional, Integral and Derivative Control Modes

Most Closed loop Controllers are capable of controlling with three control modes which can be used separately or together

- Proportional Control (P)
- Integral, or Reset Control (I)
- Derivative, or Rate Control (D)

The purpose of each of these control modes is as follows:

***Proportional Control ...***
This is the main and principal method of control. It calculates a control action proportional to the ERROR (ERR) . Proportional control cannot eliminate the ERROR completely.

***Integral Control ...(Reset)***
This is the means to eliminate the remaining ERROR or OFFSET value, left from the Proportional action, completely. This may result in reduced stability in the control action.

***Derivative Control ...(Rate)***
This is sometimes added to introduce dynamic stability to the control LOOP.

Note:
The terms ***RESET*** for integral and ***RATE*** for derivative control actions are seldom used nowadays.

***Derivative control has no functionality on its own.***

The only combinations of the P, I and D modes are:

- P For use as a basic controller
- PI Where the offset caused by the P mode is removed
- PID To remove instability problems that can occur in PI mode
- PD Used in cascade control; a special application
- I Used in the primary controller of cascaded systems

## 1.9 An Introduction to Cascade Control

Controllers are said to be "In Cascade" when the output of the first or Primary controller is used to manipulate the set point of another or Secondary controller. When two or more controllers are cascaded, each will have its own measurement input or PV but only the primary controller can have an independent set point (SP) and only the secondary, or the most down-stream controller has an output to the process.

Cascade control is of great value where high performance is mandatory in the face of random disturbances, or where the secondary part of a process contains an undue amount of phase shift.

The principal advantages of cascade control are:

- Disturbances occurring in the secondary loop are corrected by the secondary controller before they can effect the primary, or main, variable.
- The secondary controller can significantly reduce phase lag in the secondary loop thereby improving the speed or response of the primary loop.
- Gain variations in the secondary loop are corrected within that loop.
- The secondary loop enables exact manipulation of the flow of mass or energy by the primary controller.

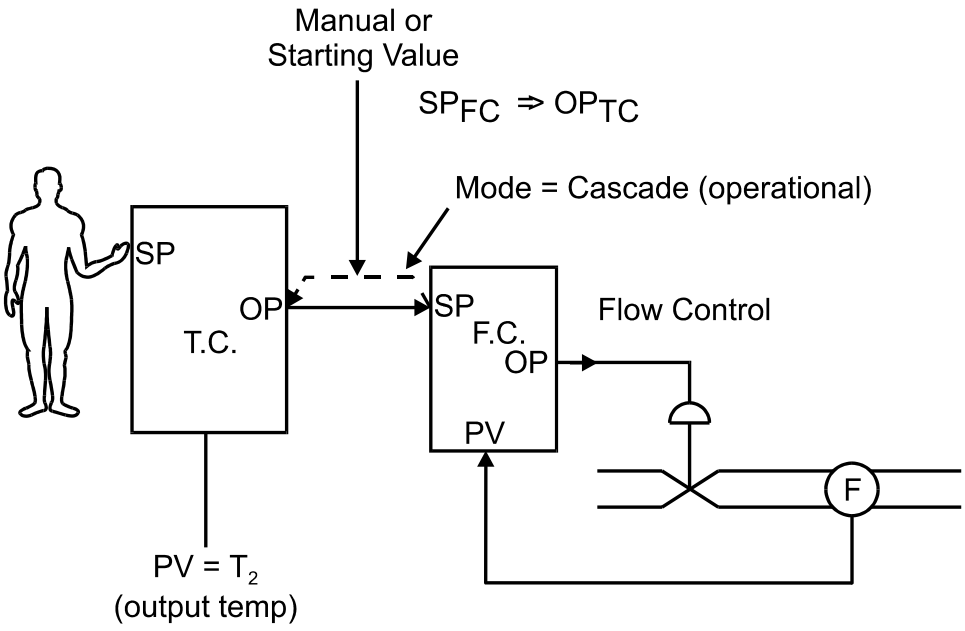



***Figure 1.12***
***An Example of Cascade Control***

An example of cascade control is shown in figure 1.12 where the primary controller, TC, is used to measure the output temperature, $T_2$, and compare this with the Setpoint value of the TC, and the secondary controller, FC, is used to keep the fuel flow constant against variables like pressure changes.

The primary controller's output is used to manipulate the SP of the secondary controller thereby changing the fuel feed rate to compensate for temperature variations of $T_2$ only. Variations and inconsistencies in the fuel flow rate are corrected solely by the secondary controller; the FC controller.